

To What Extent Did the 2023 ULEZ Expansion Affect Nitrogen Dioxide (NO_2) Concentrations in London?

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1 Motivation

Urban air pollution represents a critical public health crisis in contemporary London, with nitrogen dioxide (NO_2) concentrations linked to thousands of premature deaths annually (Greater London Authority, 2025).

In response, one of the most prominent interventions is the Ultra Low Emission Zone (ULEZ), which imposes charges on high-emission vehicles entering regulated areas (Prieto-Rodriguez *et al.*, 2022). However, Ma *et al.* (2021) provide a critical ex-post analysis, suggesting that the initial ULEZ was associated with only small improvements in air quality. More recent evidence further indicates that, despite the implementation of the ULEZ, achieving substantial improvements in London's air quality may require additional or complementary policy measures (Tong *et al.*, 2025). Together, these findings raise questions regarding the extent to which the ULEZ alone can deliver meaningful reductions in NO_2 concentrations.

This essay aims to evaluate whether the 29 August 2023 geographic expansion of the Ultra Low Emission Zone (ULEZ) is associated with a measurable reduction in NO_2 concentrations in areas newly brought under the policy. To identify this effect, I use a high-frequency dataset from the UK Automatic Urban and Rural Network (AURN)¹, comprising over 80,000 daily observations. My identification strategy relies on a Difference-in-Differences (DID) framework, comparing treated monitoring stations within London against a non-London control group, while controlling for station-specific and date-specific fixed effects to isolate the policy shock from seasonal and nationwide trends.

Empirical results indicate that, under baseline OLS and time fixed effects specifications, the ULEZ expansion is not associated with a statistically significant change in NO_2 concentrations. Under the full Difference-in-Differences model with station and date fixed effects, the estimated treatment effect of the ULEZ expansion is small in magnitude and

¹Data sourced from the UK Automatic Urban and Rural Network (AURN). Available at: <https://uk-air.defra.gov.uk>

statistically insignificant. Event-study and placebo tests provide evidence relevant to the identification strategy, while indicating limited evidence of an immediate or sustained reduction in NO₂ levels during the sample period.

2 Data and Treatment Definition

This study uses air quality monitoring data from the UK Automatic Urban and Rural Network (AURN), accessed via the UK-AIR data archive managed by the Department for Environment, Food & Rural Affairs (DEFRA). The AURN is the UK’s official air quality monitoring network and provides nationally standardised and comparable measurements of air pollutants collected from automatic monitoring stations across the country.

The analysis focuses on nitrogen dioxide (NO₂), a key traffic-related pollutant and the primary target of the Ultra Low Emission Zone (ULEZ) policy. NO₂ concentrations are measured in micrograms per cubic metre ($\mu\text{g}/\text{m}^3$). Daily mean values are used in order to reduce short-term measurement noise and to align with standard practices in policy evaluation.

The initial dataset contains daily NO₂ observations from all automatic monitoring stations included in the AURN network. The sample period spans from 1 August 2022 to 28 February 2024. This window provides a pre-treatment period prior to the London-wide expansion of the ULEZ on 29 August 2023, as well as a post-treatment period during which the policy was in effect.

Observations with missing NO₂ measurements are excluded from the analysis. Nationwide pollution shocks are absorbed by date fixed effects, ensuring that identification relies on differential within-station changes over time rather than aggregate trends. After applying these restrictions, the final dataset consists of 84,837 station-day observations, forming a panel structure in which each monitoring station is observed repeatedly over time.

Greater London consists of 32 London boroughs and the City of London. However, the empirical analysis is constrained by the spatial distribution of air quality monitoring stations and data availability. Not all London boroughs operate continuous monitoring stations included in the national air quality dataset, and some boroughs are further excluded due to incomplete NO₂ observations during the sample period. As a result, the dataset includes only those boroughs in which monitoring stations report valid NO₂ measurements throughout the study period.

Following data cleaning and sample restrictions, the treated group comprises monitoring stations located in nine Greater London boroughs affected by the August 2023 ULEZ expansion (see Table 1): Bexley, Camden, Greenwich, Haringey, Hillingdon, Kensington and Chelsea, Southwark, Tower Hamlets, and Westminster.

Monitoring stations located within Greater London and included in the final sample

Table 1: Distribution of Treated Monitoring Stations across London Boroughs

Borough	Observations	Percent (%)	Cumulative (%)
Bexley	569	8.12	8.12
Camden	1,133	16.18	24.30
Greenwich	268	3.83	28.13
Haringey	1,079	15.41	43.53
Hillingdon	1,147	16.38	59.91
Kensington and Chelsea	570	8.14	68.05
Southwark	569	8.12	76.17
Tower Hamlets	572	8.17	84.34
Westminster	1,097	15.66	100.00
Total	7,004	100.00	

are defined as the treatment group, while all other stations in the final sample serve as the control group.² In the final sample, 7,004 observations (approximately 8.3%) correspond to treated London monitoring stations, while the remaining 77,833 observations (91.7%) form the control group. The London-wide expansion of the ULEZ represents an externally imposed policy shock that affects treated monitoring stations but not control stations in the final sample, thereby providing a plausibly exogenous policy change, conditional on the parallel trends assumption.

The outcome variable is the daily mean concentration of nitrogen dioxide (NO_2). A post-treatment indicator ($PostULEZ_t$) is constructed that takes the value of one for dates on or after 29 August 2023, when the London-wide ULEZ expansion came into effect, and zero otherwise. The DID estimator exploits within-station changes in NO_2 concentrations before and after the policy, relative to contemporaneous changes observed at untreated stations.

Air quality monitoring sites differ in their local environments, such as roadside versus background locations, and individual stations may not fully represent city-wide pollution levels. However, the empirical analysis exploits within-station variation over time rather than cross-sectional comparisons across locations (e.g. Ma *et al.*, 2021). As a result, the empirical strategy helps to account for time-invariant site-specific characteristics, thereby mitigating concerns related to permanent differences across monitoring sites.

²Formally, the treatment indicator $London_i$ equals 1 if monitoring station i is located within Greater London and included in the final sample, and 0 otherwise. For expositional simplicity, these are referred to as “London” and “non-London” throughout the essay.

Table 2: Summary Statistics for Key Variables

Variable	Mean	Std. Dev.	Min	Max	N
NO_2 concentration ($\mu g/m^3$)	17.51	12.38	0	99	84,837
Treated station (London)	0.08	0.28	0	1	84,837
Post-policy period	0.32	0.47	0	1	84,837

Note: The sample consists of daily observations from monitoring stations.

3 Econometric Methodology

3.1 Baseline Model

Baseline OLS

$$NO2_{it} = \alpha + \beta (London_i \times PostULEZ_t) + \varepsilon_{it} \quad (1)$$

This model provides a baseline ordinary least squares (OLS) estimate of the association between the London-wide expansion of the Ultra Low Emission Zone (ULEZ) and nitrogen dioxide (NO_2) concentrations. The interaction term $London_i \times PostULEZ_t$ captures the difference in average NO_2 levels between monitoring stations located within London and non-London after the policy implementation. This model does not control for unobserved station-specific heterogeneity or common time shocks and therefore serves as a simple benchmark for comparison.

Time Fixed Effects

$$NO2_{it} = \alpha + \delta_t + \beta (London_i \times PostULEZ_t) + \varepsilon_{it} \quad (2)$$

This model extends the baseline OLS model by incorporating time fixed effects, denoted by δ_t , to control for common shocks affecting all monitoring stations over time. These include seasonal patterns, weather conditions, and nationwide trends in air pollution. By absorbing these aggregate time effects, the estimated coefficient on the interaction term $London_i \times PostULEZ_t$ reflects differences between London and non-London monitoring stations net of common temporal shocks. However, this specification continues to rely on cross-sectional differences across locations and does not control for time-invariant station-specific characteristics. As a result, it only could be interpreted as an intermediate benchmark.

Difference-in-Differences

To estimate the effect of the ULEZ expansion on NO_2 concentrations, I estimate the following difference-in-differences specification:

$$NO2_{it} = \alpha_i + \delta_t + \beta (London_i \times PostULEZ_t) + \varepsilon_{it} \quad (3)$$

where $NO2_{it}$ denotes the daily mean concentration of nitrogen dioxide measured at monitoring station i on day t . $London_i$ is an indicator variable equal to one if monitoring station i is located within London and zero otherwise. $PostULEZ_t$ is a post-treatment indicator equal to one for dates on or after 29 August 2023, when the ULEZ expansion came into effect, and zero otherwise. The interaction term $London_i \times PostULEZ_t$ captures the differential change in NO_2 concentrations at London monitoring stations following the policy intervention.

The specification includes monitoring-station fixed effects (α_i), which control for all time-invariant characteristics of each station, such as location, surrounding land use, and whether the station is located at the roadside or in an urban background environment. Date fixed effects (δ_t) are also included to account for common shocks affecting all monitoring stations on a given day, including weather conditions, seasonal patterns, and nationwide trends in air pollution. Under the parallel trends assumption, the coefficient β captures the average treatment effect on the treated (ATT) of the ULEZ expansion on NO_2 concentrations at treated monitoring stations.

$$\beta = \mathbb{E}[NO2_{it}(1) - NO2_{it}(0) \mid London_i = 1, PostULEZ_t = 1], \quad (4)$$

where $NO2_{it}(1)$ and $NO2_{it}(0)$ denote the potential daily NO_2 concentrations at monitoring station i on day t with and without the ULEZ expansion, respectively, with the latter representing the counterfactual outcome in the absence of the policy.

3.2 Identification Strategy

A key identifying assumption underlying the DID design is that, in the absence of the policy intervention, treated and control locations would have followed parallel trends in NO_2 concentrations. Figure 1 plots the monthly average NO_2 concentrations for treated (London) and control (non-London) monitoring stations. The figure shows that prior to the August 2023 ULEZ expansion, the two groups exhibit broadly similar pre-treatment trends, providing evidence relevant to the parallel trends assumption.

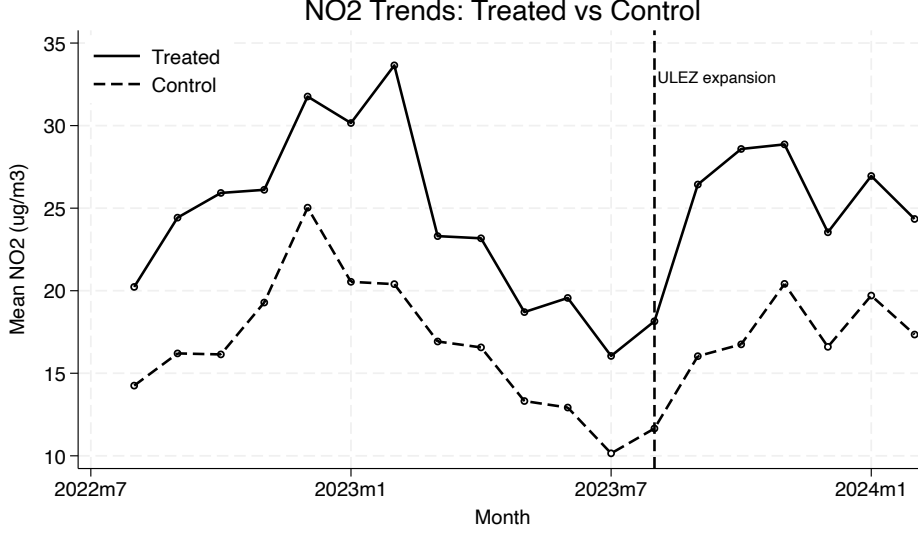


Figure 1: Monthly average NO₂ concentrations in London (treated) and non-London (control) areas

4 Empirical Results

4.1 Baseline Results

Following Bertrand et al. (2004), standard errors are clustered at the monitoring-station level throughout all specifications to allow for serial correlation within stations over time. In air pollution panel data, unobserved local factors and the persistence of emissions imply that regression residuals are correlated within stations across periods. Ignoring this dependence would lead to downward-biased standard errors and invalid statistical inference.

Baseline OLS Results

The baseline OLS results (Figure 2) show a raw association between the *ULEZ* expansion and NO₂ levels. The estimate for the interaction term (London $\times$ After Expansion) is 1.086, with a standard error of 0.874. Although the coefficient is positive, it is statistically insignificant at 10% levels ($p = 0.216 > 0.1$). Furthermore, the model reveals substantial pre-existing differences, as monitoring stations in London have NO₂ concentrations approximately $7.55 \mu\text{g}/\text{m}^3$ higher than those in the control group on average ($p < 0.01$). The low R^2 of 0.0336 suggests that this simple linear model explains only a small fraction of the total variation in pollution, likely because it fails to account for station-specific characteristics and aggregate time trends. These results highlight the necessity of introducing fixed effects to control for unobserved station-level heterogeneity and obtain a more reliable estimate of the policy effect estimate.

Linear regression

Number of obs = 84,837
F(3, 159) = 17.13
Prob > F = 0.0000
R-squared = 0.0336
Root MSE = 12.168

(Std. err. adjusted for 160 clusters in station_id)

no2_ugm3	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
treat London	7.548336	2.041439	3.70	0.000	3.516501	11.58017
post After Expansion	1.315954	.2368921	5.56	0.000	.8480935	1.783815
treat#post London#After Expansion	1.085537	.8738381	1.24	0.216	-.6402894	2.811364
_cons	16.44318	.6420951	25.61	0.000	15.17504	17.71131

Figure 2: Baseline OLS Estimates of the ULEZ Expansion

Time Fixed Effects Results

Figure 3 introduces time fixed effects to control for common temporal shocks affecting all monitoring stations, such as seasonality and meteorological conditions. As expected, the post-expansion indicator is absorbed by the time fixed effects and therefore omitted from the specification.

After controlling for time fixed effects, monitoring stations in London continue to exhibit significantly higher NO_2 concentrations than those in the control group, with an estimated difference of approximately $7.5 \mu g/m^3$ ($p < 0.01$). In contrast, the coefficient on the interaction term between London and the post-expansion period remains positive but statistically insignificant, with a magnitude similar to that obtained in the baseline specification.

Taken together, these results suggest that the absence of a detectable policy effect is unlikely to reflect common time trends or seasonal variation. However, the persistent difference between London and control locations points to substantial time-invariant spatial heterogeneity across monitoring stations. The next specification therefore adopts a difference-in-differences framework with station fixed effects, allowing identification to rely on within-station variation over time.

HDFE Linear regression	Number of obs	=	84,837
Absorbing 1 HDFE group	F(2, 159)	=	7.43
Statistics robust to heteroskedasticity	Prob > F	=	0.0008
	R-squared	=	0.3562
	Adj R-squared	=	0.3518
	Within R-sq.	=	0.0455
Number of clusters (station_id) =	Root MSE	=	9.9654
(Std. err. adjusted for 160 clusters in station_id)			

no2_ugm3	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
treat London	7.515374	2.048653	3.67	0.000	3.469293	11.56146
post After Expansion	0	(omitted)				
treat#post London#After Expansion	1.107104	.9037818	1.22	0.222	-.6778614	2.89207
_cons	16.8621	.6438938	26.19	0.000	15.59041	18.13378

Figure 3: Time Fixed Effects Estimates of the ULEZ Expansion

Difference-in-Differences Results and Robustness Checks

Table 3 reports the estimated effect of the *ULEZ* expansion on daily NO_2 concentrations under alternative specifications. Column (1) presents a simple OLS estimate without fixed effects, while Columns (2)–(3) progressively introduce time fixed effects and monitoring-station fixed effects. Column (4) reports a placebo test using a fictitious policy implementation date.

The coefficient of interest, $ULEZ \times Post$, represents the estimated DID effect (β) of the expansion on NO_2 concentrations. In the full specification (Column 3), the estimate is 0.116 with a standard error of 0.492, indicating that the result is not statistically significant at conventional levels ($p > 0.1$). Notably, the point estimate declines sharply from 1.107 in the time fixed effects specification to 0.116 once monitoring-station fixed effects are introduced. This change highlights the sensitivity of the estimated association to the inclusion of station fixed effects, which capture time-invariant differences across monitoring locations.

The placebo test in Column (4) is conducted to assess the parallel trends assumption. By shifting the treatment window to one year prior to the actual expansion (August 2022), we test whether any pre-existing systematic differences in NO_2 trends existed between London and the control group. The statistically insignificant coefficient of $Placebo \times Post$ ($\beta = 0.777$, $p > 0.1$) provides evidence that our baseline results are not reflective of pre-treatment divergent trends or unobserved annual shocks, thereby strengthening the interpretation of our main model.

Table 3: The Impact of ULEZ Expansion on NO_2 Concentrations

	(1) OLS	(2) Time FE	(3) Full FE	(4) Placebo
ULEZ $\times$ Post	1.086 (0.874)	1.107 (0.904)	0.116 (0.492)	
Placebo $\times$ Post				0.777 (0.621)
Time FE	No	Yes	Yes	Yes
Station FE	No	No	Yes	Yes
Observations	84,837	84,837	84,837	57,970

Standard errors in parentheses clustered at the station level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

4.2 Event Study and Dynamic Effects

To formally test the parallel trends assumption and examine the dynamic response of NO_2 concentrations to the ULEZ expansion, I employ an event study framework. Figure 4 plots the estimated coefficients for the eight weeks leading up to and following the policy implementation on August 29, 2023.

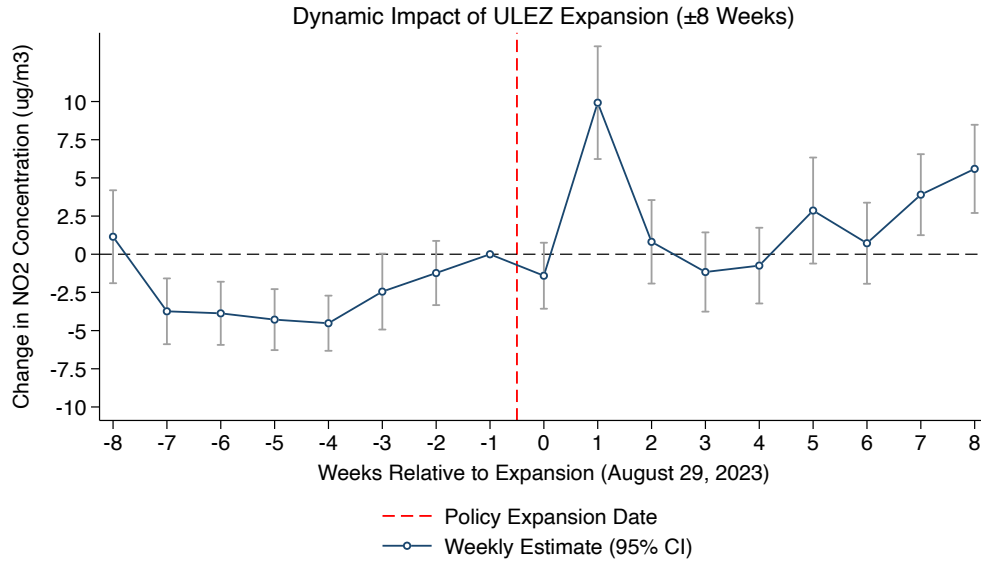


Figure 4: Dynamic impact of the ULEZ expansion on NO_2 concentrations: event-study estimates for the eight weeks before and after August 29, 2023.

The results provide evidence relevant to the identification strategy. Specifically, the coefficients for the pre-treatment period (weeks -8 to -1) are generally small in magnitude. While some minor anticipatory fluctuations are observed around weeks -7 to -4 , the lack of a systematic pre-existing trend suggests that the treatment and control

groups were following similar trajectories prior to the intervention.

Following the expansion (week 0 onwards), the estimates exhibit some volatility. There is a noticeable but transient spike in NO_2 levels immediately after the implementation in week 1. As documented by Davis (2008), driving restrictions do not necessarily translate into an immediate reduction in overall vehicle use. Instead, the policy may prompt substantial intertemporal substitution, with driving activity shifting toward hours and days when restrictions are not in place, such as nights and weekends. In the short run, these behavioural adjustments can alter the temporal distribution of traffic and emissions. When air pollution is measured at a coarser temporal resolution, such as weekly averages, these short-run reallocations may manifest as transient increases in observed pollution levels, even in the absence of sustained changes in total emissions. Moreover, as emphasized by Schlenker and Walker (2016), short-term air quality outcomes are highly sensitive to idiosyncratic weather and congestion shocks, which are difficult to fully control for using standard time fixed effects, further contributing to transient volatility in early post-treatment estimates.

However, the coefficients subsequently stabilize near zero in the following weeks. This dynamic pattern is consistent with the statistically insignificant average treatment effect reported in the baseline results, suggesting limited evidence of immediate and sustained reduction in pollution within the short eight-week window analyzed.

4.3 Robustness Checks

To further assess the robustness of our baseline findings and ensure that the results are not attributable to unobserved random factors or specific sample selection, I conduct a placebo test using a permutation approach. In this exercise, I randomly assign the treatment status (being located in London) to different monitoring stations in the sample and re-estimate the baseline DID model. This process is repeated 500 times to generate a distribution of placebo coefficients.

Figure 5 illustrates the results of this exercise. The kernel density plot shows the distribution of these 500 placebo estimates, which is approximately centered around zero and follows a clear normal distribution. The red dashed vertical line represents the actual estimated coefficient (0.116) from our baseline model in Table 3.

As illustrated in Figure 5, the vertical red dashed line representing the baseline estimate of 0.116 lies close to the zero benchmark and falls well within the main mass of the placebo distribution. While the placebo distribution exhibits some asymmetry, the estimated treatment effect does not stand out relative to the range of coefficients generated under random treatment assignment. This pattern indicates that the estimated effect is not unusual when compared with outcomes produced by random reassignment.

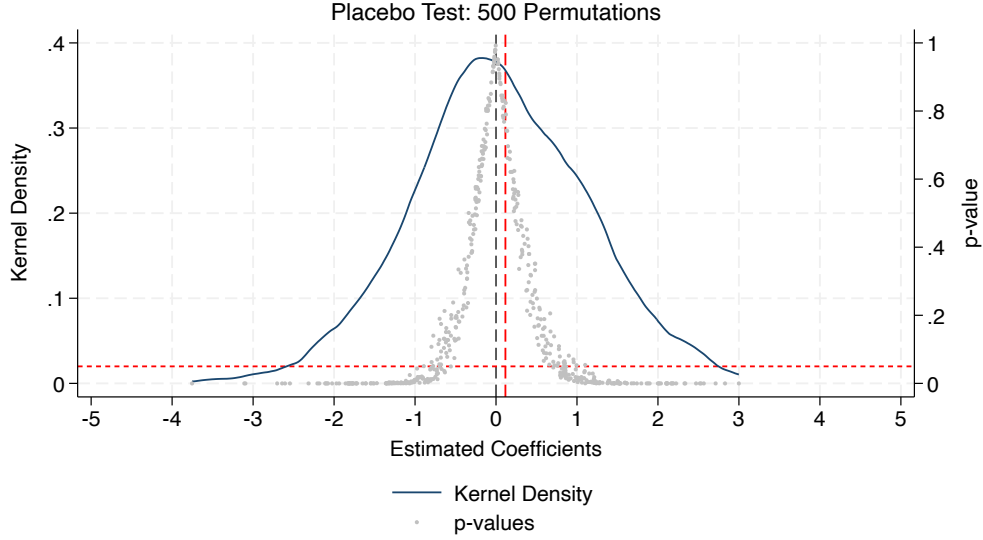


Figure 5: Placebo test based on 500 random permutations. The distribution of estimated coefficients and corresponding p -values is shown. The red dashed vertical line indicates the actual estimated coefficient, and the horizontal dashed line marks the $p = 0.05$ significance threshold.

Furthermore, the horizontal dashed line marks the conventional 5% significance threshold ($p = 0.05$). The vast majority of the 500 simulated permutations yield p -values well above this threshold, indicating that random reassignment of treatment status is rarely associated with statistically significant effects.

Overall, this placebo exercise provides additional support for the credibility of the identification strategy by illustrating that the estimated coefficient lies well within the distribution of placebo estimates.

5 Conclusion

This essay evaluates whether the August 2023 London-wide expansion of the Ultra Low Emission Zone (ULEZ) is associated with a reduction in nitrogen dioxide (NO_2) concentrations in newly treated areas. Using high-frequency monitoring data and a Difference-in-Differences framework with station and date fixed effects, the analysis finds no statistically significant evidence of an immediate or sustained decline in NO_2 levels following the policy expansion. Across baseline OLS, time fixed effects, and full DID specifications, the estimated treatment effects are small in magnitude and statistically insignificant.

Event-study and placebo tests provide evidence relevant to the identification strategy, indicating that treated and control locations followed similar trends prior to the expansion and that the null results are unlikely to reflect spurious correlations. These findings suggest that the environmental impact of the ULEZ expansion may not be detectable

over the short post-treatment window considered in this study. Future research could examine longer time horizons, alternative pollutants, or complementary policy measures to assess whether the effects of the ULEZ emerge gradually over time.

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